

Biology All In One (BAIO) Learning System

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Chapter 9: Biotechnology – Principles and Processes

Worksheet · Introduction to Biotechnology & Genetic Engineering

I. Multiple Choice Questions

- EFB stands for:
(A) European Federation of Biotechnology
(B) European Fund for Biology
(C) Environmental Federation of Biology
(D) European Foundation for Bioremediation
- Which of these is NOT listed as a direct application of biotechnology in the chapter?
(A) Agriculture
(B) Bioremediation
(C) Space exploration
(D) Waste treatment
- Genetic engineering involves introduction of foreign _____ into a host's genome.
(A) Proteins
(B) DNA/RNA
(C) Lipids
(D) Carbohydrates
- Aseptic techniques maintain what kind of environment for bioprocesses?
(A) Contamination-free (sterile)
(B) Highly acidic
(C) Anaerobic only
(D) Radioactive
- Which type of reproduction preserves genetic variation between generations?
(A) Asexual reproduction
(B) Sexual reproduction
(C) Vegetative propagation
(D) Budding

II. Fill in the Blanks

- The _____ defines biotechnology as the integration of natural science and organisms/cells/parts for products and services.
- Genetic engineering introduces foreign _____ into a host's genome, altering its phenotype.
- _____ reproduction preserves genetic information, while _____ reproduction introduces variation.
- Plant and animal hybridisation often introduces _____ genes along with desirable ones.
- The process of forming multiple identical copies of DNA is called _____.
- Foreign DNA replicates in a host only if it integrates with an _____.

III. Match the Following

Column A	Column B
1. Asexual reproduction	a. Introduction of foreign DNA/RNA into a host
2. Sexual reproduction	b. Sterile, contamination-free maintenance
3. Aseptic technique	c. Introduces variation between generations
4. Genetic engineering	d. Preserves genetic information faithfully

IV. True / False

1. Genetic engineering includes creation of recombinant DNA, gene cloning, and gene transfer. [True / False]
2. Sexual reproduction preserves genetic information without any variation. [True / False]
3. Cloning refers to the formation of a single, unique DNA molecule. [True / False]
4. Aseptic techniques maintain a contamination-free environment for bioprocesses. [True / False]
5. Hybridisation is always more precise than genetic engineering. [True / False]

V. 2 Marks Questions

1. Define biotechnology as per the European Federation of Biotechnology (EFB).

2. Why does traditional hybridisation often introduce undesirable genes into offspring?

VI. 3 Marks Questions

1. List the three basic steps involved in genetically modifying an organism.

2. Explain why foreign DNA does not always replicate inside a host cell.

VII. 5 Marks Questions

1. Explain the complete process of cloning, from the introduction of foreign DNA into a host cell to the formation of multiple identical copies. Support your answer with a simple flow diagram.
